

SAP API Classification

C&F Agent Interface · Shree Cement / Bangur

Document reference	CNF-API-CLASS-001
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Prepared by	SAP Backend Integration Team
Object	The API inventory and the standard / custom nature of each
Companion objects	CNF_API_Classification.xlsx · CNF_Delivery_Planning.xlsx (effort, validation, rules, assumptions) · API_SPECIFICATION_PROPOSAL.pdf (full specification)

Basis of preparation. Prepared without SAP system access. SAP object names, BAPI signatures and released-API references are **candidates** derived from standard SAP release knowledge. Each classification is an engineering estimate pending system verification; the checks that convert each estimate into a verified fact are maintained in the Delivery Planning object.

1 Classification framework

Class	Meaning	ABAP work involved
S Standard	A delivered SAP API, BAPI or released service performs the business operation.	Thin service wrapper: protocol mapping, authorization, idempotency, error shaping. No business logic.
C Custom over standard	No single standard API returns the required shape, but every underlying source is standard.	Composite CDS view or application class over standard reads. Logic limited to derivation and aggregation.
X New construction	The required behaviour has no standard equivalent.	Designed and built: orchestration, state management, pre-document simulation, external coordination.

A class **S** rating does not mean no development — it means the *business operation* is delivered and the remaining work is integration scaffolding. Class **X** is where schedule and defect risk concentrate.

2 The API list

API-01 to API-09 correspond one-to-one with the client process register. API-10 is a proposed addition: Create DI has no amend path without it, and a delivery must be correctable before goods issue.

ID	API	Operation	Class	Primary candidate standard source	Custom work required
API-01	Check MIGO	Read	C	Order, delivery and material document reads plus document flow	Cross-document composite; tri-state status; pending-quantity derivation
API-02	Submit MIGO	Command	S	BAPI_GOODSMVT_CREATE / API_MATERIAL_DOCUMENT_SRV	Idempotency; error shaping. MRN is Datasphere-owned, not derived here

ID	API	Operation	Class	Primary candidate standard source	Custom work required
API-03	Create DI	Command	S	BAPI_OUTB_DELIVERY_CREATE_SLS or _ST0	Logistics attribute persistence; source-document strategy
API-04	Stock Availability	Read	C	Stock and batch reads plus availability check	Composite view; allocation ranking; allocatable determination
API-05	Shipment Cost	Read + Estimate	X / S	Shipment costing for posted cost. No standard pre-document estimate	Entire pre-commitment estimate mechanism
API-06	Invoice Creation	Orchestration	X	Standard components per stage. No standard orchestration	Orchestration, state model, resumption, process idempotency
API-07	Invoice Correction	Command	S	Invoice Correction Request (RK) / BAPI_BILLINGDOC_CANCEL1	Statutory window guard; routing between correction and reversal
API-08	E-Invoice Correction	Command	S	Document Compliance — IRN cancel and reissue	Window evaluation; branch to credit note after window; E-Way linkage guard
API-09	E-Way Bill Extension	Command	S / X	Document Compliance for India	Near zero if licensed; substantial if not
API-10	Modify DI (proposed)	Command	S	BAPI_OUTB_DELIVERY_CHANGE	Editable-scope enforcement

Distribution: 5 Standard · 2 Custom over standard · 2 New construction · 2 conditional on Document Compliance licensing (API-08, API-09).

3 Standard versus custom — per API

ID	API	Class	What standard SAP provides	What must be built custom	Note
API-01	Check MIGO	C	All underlying document reads	Cross-document composite, tri-state receipt status, pending-quantity derivation	Build over released views, not raw table selects, where the release allows
API-02	Submit MIGO	S	The goods-receipt posting in full	Idempotency guard, plant authorization, error shaping	On S/4 the standard OData service may serve this directly
API-03	Create DI	S	Delivery creation in full	Logistics attribute persistence, source-document selection strategy	Vehicle and driver have a standard home on the shipment document
API-04	Stock Availability	C	Stock reads, availability check, batch master	Composite view, allocation ranking, allocatable determination, expiry exclusion	No single standard API returns the combined payload
API-05	Shipment Cost	X / S	Posted cost retrieval (ACTUAL)	The entire pre-document estimate (ESTIMATE)	Standard costing is document-bound; no pre-commitment mechanism exists
API-06	Invoice Creation	X	Every individual stage operation	Orchestration, state model, resumption, per-stage guards, concurrency, reconciliation	Four SAP units of work plus two external calls; cannot be one LUW
API-07	Invoice Correction	S	Correction request document type and cancellation	Statutory window guard, routing between two distinct processes	A memo request is a sales document, not a billing document
API-08	E-Invoice Correction	S	IRN cancellation, status and reissue via Document Compliance	Window evaluation and branch logic; E-Way linkage guard	A generated IRN cannot be amended — cancel-and-reissue inside the window, credit/debit note outside
API-09	E-Way Bill Extension	S / X	E-document lifecycle and statutory connectivity, if licensed	Near nothing if licensed; a full statutory client if not	Largest single swing in the estimate
API-10	Modify DI	S	Delivery change in full	Editable-scope enforcement	Outbound-delivery change structures, not the inbound family

Two facts frame this classification. First: API-05 ESTIMATE and API-06 account for the substantial majority of genuine construction; five of the ten APIs are standard operations needing integration scaffolding only. Second: API-08 and API-09 are both conditional on **SAP Document Compliance for India** being licensed and configured — one

licensing question moves both APIs between near-standard and new construction, the largest single swing in the inventory.

Cross-system boundary. S/4HANA is not the sole source of data for this product. The portal's pending-order, deliveries and invoice views are served from Commerce Cloud T1; depot, storage-location and related masters live in Commerce T2; MRN, STO and ageing are Datasphere-owned. This inventory covers the S/4HANA APIs only, and no API here claims authority over a value it merely displays. The full ownership model, effort mapping, business rules, validation register and assumption log are maintained as separate objects in **CNF_Delivery_Planning.xlsx**.